Smart Tourist Safety Monitoring & Incident Response System using Al ,Geo-Fencing, and Blockchain-based Digital ID

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ABSTRACT

The safety and security of tourists has been a concern in the world as tourism is one of the rapidly expanding industries. Current tourism safety mechanisms are primarily centered on tracking of their locations and emergency warnings, yet they lack intelligent risk forecasting and protection of tourist data. The following paper suggests a Smart Tourist Safety and Monitoring System which will incorporate Global Positioning System (GPS), geo-fencing, Artificial Intelligence (AI), Internet of Things (IoT), and blockchain technology. This system monitors the position of the tourist and identifies as abnormal behavior, risks forecasting through artificial intelligence algorithms, and sending warnings when a tourist is outside of the safe areas within the limits. IoT devices gather real-time environmental and health related data, whereas blockchain guarantees safe and impeccable storage of tourist information. The suggested model enhances the safety of tourists, helps decrease false alarms, and respond to the emergency more quickly. The system is scalable and lightweight and can be used in smart tourism.

Keywords Smart Tourism, Tourist Safety, GPS Tracking, Geo-Fencing, Artificial Intelligence, Internet of Things, Blockchain

I. INTRODUCTION

Tourism has a great impact on world economy. However, the problems that tourists commonly face are some loss, theft, accidents, health related problems, and insecure destinations. The most common classic tourist safety systems are based on manual support, emergency call, or simple tracking on a GPS. These plans are insufficient because they fail to foresee the harmful situations even before they occur.

The combination of modern technologies (AI, IoT, GPS, blockchain) can be implemented in order to develop an intelligent tourist monitoring system. GPS makes it possible to track in real-time, geo-fencing is used to create safe zones, AI is able to predict dangerous circumstances, IoT gathers sensor data and blockchain secures confidential data.

II. LITRATURE SURVEY

The literature review indicates that various researchers have been engaged in enhancing tourist safety with the help of various technologies. The early tourist surveillance systems depended mostly on GPS tracking that ensured that the whereabouts of the tourists was always known by them and they could be helped in case of an emergency. These systems might assist in real time tracking, which was quite useful, but could not predict any potential hazard prior to its occurrence. The researchers also suggested the geo-fencing methods to provide protective protection of the tourist regions. An alert was produced when a tourist was out of the stipulated area. Even though geo-fencing enhanced safety, it resulted in frequent cases of false alarms since it only verified the location of the tourist, but not the actions nor circumstances of the tourist.

The recent studies were applied to the sphere of Artificial Intelligence and Machine Learning to detect abnormal tourist behavior and predict risk. Artificial intelligence systems monitor movement, velocity, the history of locations, and the environment to identify whether a tourist is at risk or not. Such techniques offer more precision as compared to conventional systems. However, the majority of these AI models include huge data and heavy calculations, and therefore they are not convenient to operate in real-time and in low-priced devices. The Internet of Things (IoT) technology has also promoted the safety of tourists. Wearable devices or mobile app IoT sensors can gather information on the body temperature, heart rate, movement, and the environment. This data assists the system to track the status of the tourist at all times. However, most of the existing IoT-

based tourism systems lack strong security and smart prediction features and merely gather the data.

According to the overview of the literature on the matter, it is clear that the existing systems are generally based on one or two technologies, which can be GPS, IoT, AI, or blockchain. There are only a few systems that incorporate all these technologies under one platform. Thus, the suggested work is a combination of GPS, geo-fencing, AI, IoT and blockchain that will establish a comprehensive tourist safety and monitoring system. The system suggested by the authors solves the weaknesses of the other studies which suggest a system that provides real time monitoring, predicts risks in an intelligent fashion, provides fewer false alarms, can run on the lightweight computers and stores the information safely.

.III .PROPOSED SYSTEM

The proposed system is a Smart Tourist Safety and Monitoring System that is built on the combination of GPS, geo-fencing, Artificial Intelligence (AI), Internet of Things (IoT), and blockchain technology to guarantee a comprehensive safety and security of a tourist. This system starts as the tourist enters personal details, phone contacts in the event of an emergency, and travel details via a mobile application or a wearable device. The visitor will have a smartphone or wearable device that is connected to the system once they have been registered. By attaching a GPS unit to the position of the tourist, we will be able to keep track of the location of the tourist then relay the position of the tourist to the monitoring system. This provides the family members the travel agencies or even the emergency authorities with an idea on the current position of the tourist at any particular time. Real time implies that it is applicable especially in situations when a tourist gets lost or is in a crisis.

The geo-fencing module constructs virtual fences in safe and non-safe areas of tourists. Hotels, tourist areas, hiking trails, and routes used during the travel are safe and unsafe areas, whereas forests, restricted areas, lonely areas, and dangerous areas are unsafe areas. The system identifies the change each time the tourist leaves the safe area or goes to unsafe area. Compared to the traditional geo-fencing system where an alert is triggered when the tourist enters a particular site, the proposed system will also scan the behavior and the movement pattern of the tourist and subsequently an alert is given. This reduces the false alarm or absence of it and boosts the system credibility.

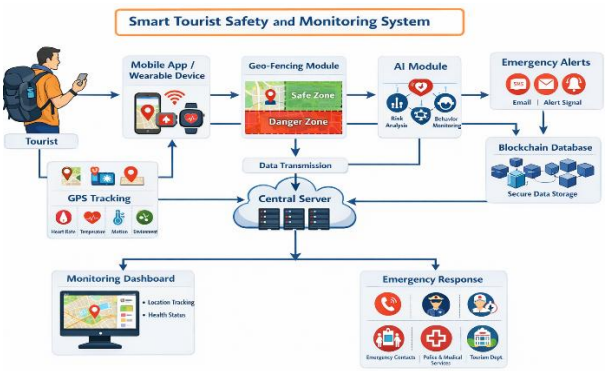
The largest element of the suggested system is the one of Artificial Intelligence as it predicts the threat and identifies the existence of abnormal cases. GPS, geo-fencing, and IoT sensors can provide the AI module with data and determine the movement, speed, direction of movement, body condition, and environmental factors of the tourist. Otherwise, one can refer to scenarios when the AI system identifies dangerous situations in the tourist as halting abruptly in a forest area, remaining idle over a long period, getting into a dangerous area, or being characterized by uncharacteristic parameters of the body, including high temperatures or unusual heart rate. The system then takes the risk score and denotes the condition as a low risk condition or a medium risk condition or high risk condition. A low hazards case points to the fact that the tourist is safe and moves regularly. An average situation would mean that the tourist may require attention, whilst high-risk situation would mean the danger.

The IoT ion also contributes to improving the efficiency of the system structure to acquire more data concerning sensors embedded in Wearable devices or smartphones. Examples of such sensors could be temperature sensors, heart rate sensors, motion sensor, accelerator and environmental sensors. The IoT devices constantly keep checking the health of the tourist and the environment. As an example, when a tourist falls down the motion sensor or accelerometer reacts to the abrupt change. In case the tourist has queer heart beats or temperature, his or her sensors immediately identify that. Likewise, there are environmental sensors, which are able to identify unsafe weather or environmental conditions. All this data is translated to the AI unit which is further broken down and a decision is reached.

The blockchain module allows offering security and privacy to the entire system. Blockchain keeps tourist data such as identification, map track, alerts on emergencies, sensor readout and travel records. It cannot be manipulated by the ability to tamper with the stored information or abuse it since blockchain is implicitly compound-proof and decentralized. This will ensure that the personal information of the tourist is safe and secure

The general functioning of the suggested system is easy and efficient. GPS and IoT sensors continuously utilize information regarding the position of the tourist and sensors accumulate essential data about the tourist. The geo-fencing confirms the existence of the tourist within the area of safety and the data retrieved is analyzed using the AI to detect the danger. In case the system notices a threatening scenario, it forwards this instantaneously to emergency contacts, rescue services or government using SSM or mobile messages or email. Finally, but not the least, it all is safely kept in blockchain. The proposed system is therefore superior to the available tourist safety systems because it does not just track the tourist, but also recognize danger, reduce false alarms, improve emergency response in addition to a safe storage of information.

STEPS FOR THIS PROJECT/BASIC ARCHITECTURE



IV. IMPLEMENTATION PLAN:

A mobile application and a central server are developed to initiate the implementation of the proposed Smart Tourist Safety and Monitoring System. The mobile app is downloaded to the smartphone or wearable device of the tourist and is utilized in the registration, sharing his/her location, and communicating during emergencies. Under registration, the tourist puts in the personal information, emergency contacts, and travel details. All these details are stored in a database whereby the server is linked to the mobile application to deal with all information and communication among various modules.

The second stage is the implementation of GPS tracking module. Continuous assessment of the location of the tourist is done by the GPS sensor embedded in the smartphone or the wearable device. The server is frequently fed with the location coordinates. There is also a map interface which is designed in such a way that the position of the tourist can be seen visually. The system then tracks the movement of the tourist to determine whether he or she is moving normally or accessing unknown areas. This creates the fundamental foundation of real-time tourist surveillance.

The geo-fencing module is developed after installing GPS tracking. At this phase, virtual fences are drawn around the secure tourist sites and unsafe zones. The system stores these boundaries by use of longitude and latitude values. Each time the tourist crosses the specified boundary, the system determines the current position, and compares it to the limits of geo-fence. In case the tourist is transferred to a risky region, the system is ready to issue an alert. In order to minimize false alarms, the notification does not go directly to the user, the system first verifies the movement and activity of the tourist via the AI modules.

The modules of AI and IoT are then implemented simultaneously. The mobile device or wearable band is connected to the IoT sensors which may include heart rate sensors, temperature sensors, accelerators, and motion sensors. These are sensors that constantly monitor the health and movements of the tourist. Based on sample data, the AI module is trained to determine normal and abnormal tourist behavior. Parameters analyzed by the AI algorithm include sudden halting, erratic velocity, unusual body temperature, immobility or unsafe direction of its travel. According to these values, the system compares the risk score, at which a tourist is safe or dangerous.

The blockchain module is provided when the monitoring system is running well. A blockchain system is developed which stores valuable records including tourist information, location history, risk notifications and emergency messages. Each time that new data is created, it is transformed into blocks and it is appended to the blockchain. This will ensure that data cannot be changed or deleted by unauthorized parties. The stored information is only accessible to the authorized individuals including police, hospitals, travel agencies or emergency contacts. The effect of this implementation is security, privacy and trust in the system.

Lastly, the entire system undergoes testing in various tourist scenarios like in normal travelling, access to risky zones, sensor defects, and emergency cases. Measures of the response time, the accuracy of risk prediction, false alarms, and data security are registered. The necessary improvements are given on the basis of the test results. Upon effective testing, the system can be implemented in tourist destinations, pilgrimage hubs, trekking sites, smart cities, and travel agencies to enhance the safety of tourists and response to emergencies.

V. RESULTS/DISCUSSIONS:

The suggested Smart Tourist Safety and Monitoring System was administered to various tourist scenarios like normal movement, entry to unsafe areas, abrupt stopping, abnormal health and emergency cases. Through GPS, the system was in a position to track the location of the tourist in real time and easily tracked the tourist at all times with the assistance of the IoT sensors. Accordingly the geo-fencing module could detect the presence of the tourist in the safe zone. The system was alerted and provided announcements to the emergency contacts, which has been registered to the tourist, every time the latter entered a dangerous area.

The result denotes that the risk forecasting module based on AI helped to play up the work of the system. The proposed system contrasted movement pattern, speed, inactivity, body temperature, heart rate and provided an alert as opposed to the classical systems of tourist monitoring which only displays the location tracking. This intelligent analysis meant that the system would be aware of dangerous cases early enough and provide more timely warning messages. As an illustration, when the tourist was lying in one location and had not moved in a very long time in an unsafe region, the system registered this status as having a high-risk incident and it instantly alerted the emergency contacts.

The geo-fencing module was also better than the existing mechanisms as it reduced false alarms. When a tourist in a traditional system crosses the barrier, an alarm is through any means raised irrespective of whether the transit is temporary or not dangerous. This can be realized in the suggested system where the AI module will be analyzing the actions of the tourist and trigger the alarm. This means that when the

tourist loiter out of the safe area and returns without any action the system will not send unnecessary messages. This made the monitoring system reliable and correct.

An additional support given to the system was the IoT sensors used to monitor the condition of health and the environment of the tourist. During the testing process, the motion sensor and the accelerometer were successful with respect to empty movement and sudden falls. In the same manner, the temperature and heart rate monitor detected unhealthy conditions. These sensor values impacted positively on the quality of the decisions taken by the AI module and increased the accuracy of risk prediction. Through this, the system was in a position to give a warning even prior to a serious emergency.

The blockchain module was able to store all the tourist data, a history of the location, sensor measurements and emergency messages in a secure and safe way. The data stored could be neither changed nor lost in the absence of authorization, which increased the security and privacy of data. The blockchain allowed the storing of the information that could be accessed by authorized personalities such as travel agencies, police, and hospitals and used to respond to an emergency. This reduced the time delay in the rescue mission and generated facts to the reliability of the system.

Generally, the discussion on the findings reveals that the proposed system is more efficient as compared to the existing tourist safety systems. The existing systems are mostly based on GPS and geo-fencing, yet the proposed work will combine GPS, AI, IoT, and blockchain and provide the entire solution. The system has enhanced risk forecasting, emergency response, reduction of false events and maximum enhancement of data security. Therefore, the proposed Smart Tourist Safety and Monitoring System is more efficient and suitable in the context of the modern use of tourism applications.

VI. OUTPUTS:

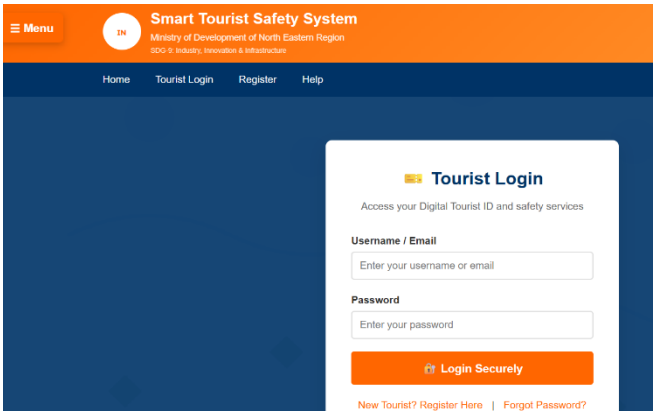


FIGURE 1.1: User Login Page.

This Application provides a secure login interface, in which users can sign in with a username and password or create a new account

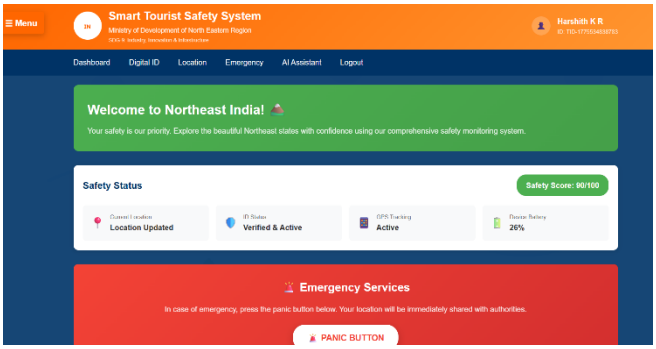


FIGURE 1.2: Dashboard Page.

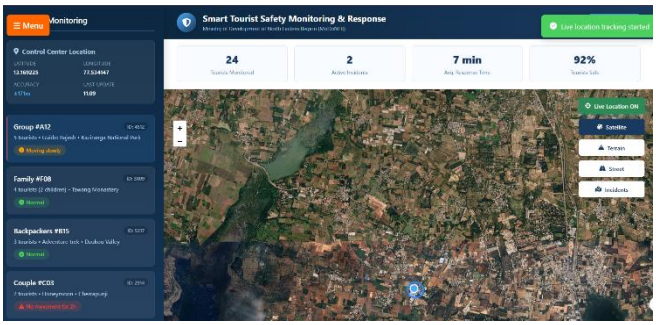


FIGURE 1.3: Live Location Monitoring Page

This page allows users to open and read the selected book directly within the platform.

VII. FUTURE WORK:

It is also planned in the future that the proposed Smart Tourist Safety and Monitoring System will be enhanced with facial recognition technology. This feature may be useful to detect tourists, children and the old among others in populated areas like railway stations, temples, airports and tourist attraction sites. Facial recognition is also applicable to identify who is a tourist and ensure that any unauthorized individual does not enter starved or dangerous zones. This will make the system more secure and applicable in the popular tourist spots.

The next valuable change in the future is the application of more efficient Artificial Intelligence and deep learning algorithms. The existing system performs simple AI techniques to identify suspicious behavior and hazardous circumstances. However, in the future, deep learning will be able to study more intricate movement patterns, travel history and health of tourists. This will assist in making the system more accurate in predicting danger and sending an early warning before an emergency actually takes place. Due to this fact, the system has the ability to decrease accidents and increase the safety of the tourists.

Cloud computing can also be used to improve the proposed system. Cloud technology has the capacity to hold large interfaces of tourist details, sensor facts, and travel records of various destinations. This will allow the system to

support many tourists at the same time without reducing performance. The system will also be more appropriate to the smart cities, huge tourist destinations, and international tourism uses by the application of cloud computing.

In the future, the system will be linked directly with the hospitals, police stations, rescue teams, travel agencies and the government tourism departments. The system has a possibility to notify the nearest authority whenever it detects the presence of a high-risk situation by automatically communicating the location of the tourist, the health and emergency information to the nearest authority. This will diminish the time delay in rescuing the tourists and this will enhance the likelihood of availing prompt assistance to the tourist.

Weather prediction, traffic tracking, crowd-sourcing, and voice-activated emergency support can also be an element of the system. Before going to a destination, tourists can be notified of a potential heavy rain, flooding, landslides, traffic jam, or congestion. Voice recognition will enable the tourist to send an emergency message by merely speaking, a feature that will come in handy in situations where the tourist is injured or has no access to the phone. This can enhance the system to be more intelligent, reliable, and helpful in future uses in tourism.

VIII. CONCLUSION:

The proposed Smart Tourist Safety and Monitoring System is a good solution to the enhancement of the safety and security of the tourists. The current systems of tourist safety are primarily oriented at how to track the location of tourists and provide them with basic alerts of an emergency, yet they cannot foresee the potential dangerous scenarios or secure tourist data in an appropriate way. The proposed system will solve these limitations by integrating GPS, geo-fencing, Artificial Intelligence, Internet of Things, and blockchain technology into one solution.

The GPS module will constantly monitor the position of the tourist whereas the geo-fencing module will verify the presence of the tourist in a secure zone. As the IoT sensors keep track of the health and the surrounding environment, all the acquired information is examined by the artificial intelligence component to determine any abnormal behavior and forecast danger. When the system identifies a dangerous situation, it can send alerts to emergency contacts/ authorities automatically. Moreover, the blockchain technology records any tourist-related data in a highly secured system and avoids hacking and alteration.

The findings presented in the suggested system indicate its superiority to the current approaches. It yields more precise forecasting of the risk, lowers false alarms, enhances responsiveness in case of an emergency, and ensures the data storage is safe. The system is portable, dependable and can be applied to various tourism activities including pilgrimage centers, trekking, adventure travels, smart cities, and women security.

Thus, one can draw a conclusion that proposed Smart Tourist Safety and Monitoring System is a new and smart method of tourist protection. Not only does it monitor the tourist but it also is able to predict danger even in advance as several technologies are combined within one system. Therefore, the suggested work can become significant in enhancing tourist safety and creating smarter tourism services in the future.

IX. ACKNOWLEDGEMENT:

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